Rayen Chemlali

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SUMMARY

Currently a student at INP Toulouse – ENSEEIHT (N7) on an academic exchange, I am completing the final year of my engineering degree, majoring in HPC & Big Data, alongside a Master's degree in the PSMSC track (Scientific Computing). I come from a 4-year Software Engineering program at INSAT (National Institute of Applied Science and Technology) in Tunisia. Passionate about mathematics and computer science, I have taken part over the course of my studies in several programming contests and hackathons, alongside gaining professional experience through internships, freelance work and part-time roles.

KEY SKILLS

- **Generative & Agentic AI:** NLP, LLMs (prompt engineering, fine-tuning, embeddings), RAG, Agentic AI, multi-agent architecture, tool calling, LangChain, LangGraph, Qdrant
- **Machine Learning & Deep Learning:** Supervised & Unsupervised Learning, CNN, Transformer Architecture, time series, audio classification
- **Scientific Computing & HPC:** parallel and distributed computing, HPC algorithms, OpenMP, MPI, CUDA, OpenCL
- **Optimization:** Operations research (LP, MILP), heuristic search algorithms (A*), Gurobi
- **Big Data:** Hadoop, Spark, Kafka
- **Programming Languages:** Python, C++, Java, Go, JavaScript, PHP
- **Frameworks:** FastAPI, NestJS, ASP.NET, Symfony, React, Next.js, Angular
- **Software Development & Tools:** Software architecture, Design Patterns, Relational and non-relational databases, Git, Docker, Kubernetes, CI/CD, DevOps

PROFESSIONAL EXPERIENCE

IRIT (RMESS team) – INP Toulouse – ENSEEIHT
AI & IoT Research Project (long team project) *October 2026 – February 2027*

- **Health Guardian – Personal AI Twin**: behavioral digital twin that learns an individual's normal patterns from connected-sensor time series and detects gradual drift.
- Anomaly and drift detection on multimodal time series, with uncertainty and explainability (ongoing study, prototype in February 2027).

Plug & Tel
ML Intern *Jun 2026 – Aug 2026*

- Reduced the classification error rate of a production AI model for answering machine detection (AMD).
- Kept latency compatible with the real-time constraints of VoIP streams.
- Explored several approaches: tabular models, CNNs on mel-spectrograms and hybrid models combining both.

Part-Time Software Engineer *Nov 2025 – Apr 2026*

- Developed VoIP microservices in Go.
- Took part in the production deployment process.

Talan Tunisie Consulting
Summer Intern *Jun 2025 – Aug 2025*

- Designed and built the Smart Container Registry end to end, an AWS ECR alternative backed by S3, to cut Docker image storage costs in Kubernetes environments.
- AI agent with dynamic tool calling for storage cost analysis.
- Unified Kubernetes/S3 dashboard and an automated image deletion rules engine. *Frontend* *Backend* *Demo*

Freelance
Software Developer *Various dates*

- **Strategy Copilot**: AI-powered business strategy copilot with a chatbot and graphical visualizations, generating diagnostics, financial projections and market analyses with PDF exports.
- **Shannon**: Co-developed a C++ code generator for electronic boards, based on graph modeling with an interactive Simulink-inspired canvas. *Demo*

EDUCATION

INP Toulouse – ENSEEIHT (N7) *2026 – 2027*
5th year, HPC Big Data track (academic exchange) and Master's degree, PSMSC track (Scientific Computing)

INSAT – National Institute of Applied Science and Technology, Tunisia *2022 – 2026*
1st to 4th year in Software Engineering

CERTIFICATIONS

- Efficient LLM Customization (NVIDIA)
- Building RAG Agents with LLMs (NVIDIA)
- Samsung Innovation Campus AI Program

LANGUAGES

French (fluent), English (fluent), Arabic (native)

ACHIEVEMENTS

- Thematic winner (Enedis track) – AI For Impact Hackathon by La Mêlée, Oct 2026
- Top 6 – Data Quest 2026, Feb 2026 *Code* *Demo*
- 3rd place – AI 4 New Generation Hackathon, EY & Univ. Ibn Khaldoun, 2024
- Bronze medal (11th) – TCPC 2024
- 2nd place – Code Quest 2023, INSAT